



Secure Packages with CodeArtifact



Abdi Ali

Developer Tools > CodeArtifact > Repositories > project-devops-cicd

project-devops-cicd info

Repository Setting up Secure Packages with CodeArtifact

Delete repository Apply repository policy Edit

► **Details**
Domain, policy, tags, ARN, and upstream repositories.

Packages info

Filter by package name prefix, format, namespace prefix, and origin controls

◁ 1 ▷ ⚙

	Package name	Namespace	Format	Latest version	Latest publish date	Publish	Upstream
☐	junit	junit	maven	3.8.1	5 minutes ago	Block	Allow



Introducing Today's Project!

In this project, I will demonstrate how to Secure Packages with Code Artifact, I'm doing this project to practice and utilize my skills as a Cloud Engineer and being able to secure packages will be a fundamental skill.

Key tools and concepts

Services I used were CodeArtifact, GitHub, EC2. Key concepts I learnt include on how to deploy packages with CodeArtifact, how to install and run web apps on EC2 and using GitHub

Project reflection

This project took me approximately 3 hours, The most challenging part was solving errors with Git and the XML file.

This project is part of a series of DevOps projects where I'm building a CI/CD pipeline!



CodeArtifact Repository

CodeArtifact is an ideal tool to use to manage dependencies as its reliable and can be controlled, it is also more secured as you download packages from it instead from other potentially unsecure sources

A domain is essentially a folder which stores multiple repositories belonging to your project or organisation for simplicity and security

A CodeArtifact repository can have an upstream repository, which is essentially a backup/redundancy to the main repository.

Repositories (2)				Delete repository	View connection instructions	Create repository
Q			< 1 > 			
	Repository name	Repository description	Repository administrator			
☐	maven-central-store	Provides Maven artifacts from Maven Central Repository.	264642482087			
☐	project-devops-cicd	Setting up Secure Packages with CodeArtifact	264642482087			



CodeArtifact Security

Issue

I ran into an error when retrieving a token (EC2 Instance must have access to the other AWS services in this case CodeArtifact.)

Resolution

To resolve the error with my security token i did some troubleshooting in the terminal with some AI assistance.

It's security best practice to use IAM roles due to the principle of least privilege and Zero Trust



The JSON policy attached to my role

The JSON policy I set up grants the EC2 instance the authorization to access the Tokens, endpoints and read rights.

codeartifact-project-consumer-policy [Info](#)

[Edit](#) [Delete](#)

Policy details

Type	Creation time	Edited time	ARN
Customer managed	July 07, 2026, 16:37 (UTC+02:00)	July 07, 2026, 16:37 (UTC+02:00)	arn:aws:iam::264642482087:policy/codeartifact-project-consumer-policy

Permissions

Entities attached

Tags

Policy versions (1)

Last Accessed

Permissions defined in this policy [Info](#)

[Copy](#) [Edit](#) [Summary](#) [JSON](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "codeartifact:GetAuthorizationToken",
8         "codeartifact:GetRepositoryEndpoint",
9         "codeartifact:ReadFromRepository"
10      ],
11       "Resource": "*"
12     },
13     {
14       "Effect": "Allow",
15       "Action": "sts:GetServiceBearerToken",
16       "Resource": "*",
17       "Condition": {
18         "StringEquals": {
```



Maven and CodeArtifact

To test the connection between Maven and CodeArtifact, I compiled my web app using settings.xml

The settings.xml file configures Maven to connect with CodeArtifact repository

Compiling means turning human readable code into code machine can understand

```
GNU nano 8.3 settings.xml
<settings xmlns="http://maven.apache.org/SETTINGS/1.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/SETTINGS/1.0.0
    https://maven.apache.org/xsd/settings-1.0.0.xsd">

  <servers>
    <server>
      <!-- This ID must match your repository ID below -->
      <id>project-project-devops-cicd</id>
      <username>aws</username>
      <password>${env.CODEARTIFACT_AUTH_TOKEN}</password>
    </server>
  </servers>

  <profiles>
    <profile>
      <id>project-project-devops-cicd</id>
      <repositories>
        <repository>
          <id>project-project-devops-cicd</id>
          <!-- Use your secure HTTPS URL here, NOT the ARN -->
          <url>https://project-264642482087.d.codeartifact.us-east-1.amazonaws.com/maven/project-devops-cicd</url>
        </repository>
      </repositories>
    </profile>
  </profiles>

  <activeProfiles>
    <activeProfile>project-project-devops-cicd</activeProfile>
  </activeProfiles>
</settings>
```



Verify Connection

After compiling, I checked the project repository and I noticed that it updated the marvin upstream repository aswell

[Developer Tools](#) > [CodeArtifact](#) > [Repositories](#) > project-devops-cicd

project-devops-cicd [Info](#)

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[Delete repository](#) [Apply repository policy](#) [Edit](#)

► Details

Domain, policy, tags, ARN, and upstream repositories.

Packages [Info](#)

[Refresh](#) [Delete package](#) [View connection instructions](#)

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